y = 3
Putting the value of y in eq^{n} (i)
y + 1 = x
3 + 1 = x
x = y
the number of flowers and bees respectivd are 3 and 4.

Directions for the following 5 (five) items : Consider the following information and answer the five item that follows :

There are five persons in a group-P, Q, R, S and T. The group has one doctor, one lawyer and one artist. P and S are unmarried students. T is a man married to one of the group members. Q is the brother of P and is neither doctor nor artist. R is not doctor .

For 15 to 19

Since Q is a brother of P, so Q must be a man .

Now , Q and T (a married man) are male

- P and S are unmarried, So T must be married to R. So, R is a female.
  P and S are unmarried students.
  Q is neither a doctor nor an artist.
  R is not a doctor
- T must be a doctor

Out of Q and R, one is a lawyer and another is an artist. And Q is not an artist, so he must be a lawyer.

∴ R is an artist.

The solution of question no. 15 to 19:-

Person			Profession
P			Unmarried student
Q			Lawyer
R			Artist
S			Unmarried Student
T			Doctor
and T	Male Married to	R	Female

15. Who is the doctor ?
(a) T (b) P
(c) Q (d) R

Ans. (a) : 'T' is the doctor.

16. Who is the artist ?
(a) P (b) Q
(c) R (d) T

Ans. (c) : 'R' is the artist.

17. Who is the spouse of R
(a) P (b) T
(c) Q (d) S

Ans. (b) : 'T' is the husband of R.

18. Who is the lawyer ?
(a) P (b) Q
(c) R (d) S

Ans. (b) : Q is the lawyer.

19. Who is the following is definitely a man ?
(a) P
(b) S
(c) Q
(d) None of the above

Ans. (c) : Since Q is the brother of P. Therefore 'Q' is definitely a man.

20. There is an order of 19000 quantity of a particular product from a customer. The firm produces 1000 quantity of that product per out of which 5% are unfit for sale. In how many days will the order be completed ?
(a) 18 (b) 19
(c) 20 (d) 22

Ans. (c) : The number of days when the order be completed $\Rightarrow \dfrac{19,000}{1000-1000\times\frac{5}{100}}$

$= \dfrac{19000}{1000-50} = \dfrac{19000}{950}$

$= 20$ days.

Direction for the following 5 (five) items : Read the following two passages and answer the items that follow each passage. Your answers to these items should be based on the passages only.

Passage-1

Biomass as fuel for power, heat, and transport has the highest mitigation potential of all renewable sources. It comes from agriculture and forest residues as well as from energy crops. The biggest challenge in using biomass residues is a long-term reliable supply delivered to the power plant at reasonable costs; the key problems are logistical constraints and the costs of fuel collection. Energy crops, if not managed properly, compete with food production and may have undesirable impacts on food prices. Biomass production is also sensitive to the physical impacts of a changing climate.

Projections of the future role of biomass are probably overestimated, given the limits to the sustainable biomass supply, unless breakthrough technologies substantially increase productivity. Climate-energy models project that biomass use could increase nearly four-fold to around 150-200 exajoules, almost a quarter of world primary energy in 2050. However the maximum sustainable technical potential of biomass resources (both residues and energy crops) without disruption of food and forest resources ranges from 80-170 exajoules a year by 2050 and only part of this is realistically and economically feasible. In addition, some climate models rely on biomass-based carbon capture and storage, an unproven technology to achieve negative emissions and to buy some time during the first half of the century.

Some liquid bio fuels such as corn-based ethanol, mainly for transport, may aggravate rather than ameliorate carbon emissions on a life-cycle basis. Second generation befouls, based on lingo-cellulose feedstock –such as straw, bagasse, grass and wood-hold the promise of sustainable production that is high-yielding and emit low levels of green house gases, but these are still in the R & D stage.

21. What is/are the present constraint/ constraints in using biomass as fuel for power generation ?
1. Lack of sustainable supply of biomass
2. Biomass production competes with food production
3. Bio-energy may not always be low carbon on life-cycle basis

Select the correct answer using the code given below
(a) 1 and 2 only
(b) 3 only
(c) 2 and 3 only
(d) 1, 2 and 3

Ans. (d) : Lack of a sustainable supply of biomass production competitive with food production and bioenergy may not always be low carbon, depending on the life cycle. The above facts are the existing constraints in using biomass as fuel for power generation.

22. Which of the following can lead to food security problem ?
1. Using agricultural and forest residues as feedstock for power generation
2. Using biomass for carbon capture and storage
3. Promoting the cultivation of energy crops

Select the correct answer using the code given below?
(a) 1 and 2 only
(b) 3 only
(c) 2 and 3 only
(d) 1, 2 and 3

Ans. (b) : The problem of food security may arise due to the promotion of the cultivation of energy crops. The main food items will not be affected due to high production of coarse grains and Jatropha in the food items, otherwise there will be problem if the main food items are used as energy crops.

23. In the context of using biomass, which of the following is/are the characteristic/characteristics of the sustainable production of biofuel.
1. Biomass as a fuel for power generation could meet all the primary energy requirements of the world by 2050.
2. Biomass as a fuel for power generation does not necessarily disrupt food and forest resources
3. Biomass as a fuel for power generation could help in achieving negative emissions, given certain nascent technologies.

Select the correct answer using the code given below :
(a) 1 and 2 only
(b) 3 only
(c) 2 and 3 only
(d) 1, 2 and 3

Ans. (c) : Considering on emerging technologies, biomass as a fuel for power generation can help in achieving negative emissions. It is characterized by sustainable production in biofuels in terms of biomass use.

24. With reference to the passage, following assumptions have been made :
1. Some climate-energy models suggest that the use of biomass as a fuel for power generation helps in mitigating greenhouse gas emissions.
2. It is not possible to use biomass as a fuel for power generation without disrupting food and forest resources

Which of these assumptions is / are valid ?
(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

Ans. (a) : Some climate energy models suggest that the use of biomass as a fuel for power generation is helpful in reducing greenhouse gas emissions. This written assumption is made in the context of this passage.

Passage–2

We are witnessing a dangerous dwindling of biodiversity in our food supply. The green revolution is a mixed blessing. Over time farmers have come to rely heavily on broadly adapted, high yield crops to the exclusion of varieties adapted to the local conditions. Monocropping vast fields with the same genetically uniform seeds helps boots yield and meet immediate hunger needs. Yet high-yield varieties are also genetically weaker crops that require expensive chemical fertilizers and toxic pesticides. In our focus on increasing the amount of food we produce today, we have accidentally put over selves at risk for food shortages in future

25. Which among the following is the most logical and critical inference that can be made from the above passage ?
(a) In our agricultural practices, we have become heavily dependent on expensive chemical fertilizers and toxic pesticides only due to green revolution .
(b) Monocroping vast fields with high-yield varieties is possible due to green revolution
(c) Monocroping with high-yield varieties is the only way to ensure food security to millions
(d) Green revolution can pose a threat to biodiversity in food supply and food security in the long run

Ans. (d) : The most logical and critical inference that comes from the above passage is that the green revolution can present a threat to biodiversity in the food security in the long run. Because the root of the green revolution is to use chemical fertilizers in large quantities which are harmful for biodiversity and also not beneficial for the long term of the land.